# Feature Engineering

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

## One readable feature-and-evidence log.

Trace areas/ratios,apply training-only centering and interpret controlled comparison limits.

**Keep one habit:** W1–W3/W5 → selected new area → project feature log → changed-case return.

MAIN LESSON · SECTION 1

## Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

### Learning Curves

> **My remembered idea, example and correction**

### Reproducible Experiments

> **My remembered idea, example and correction**

### Hold Out Test Data

> **My remembered idea, example and correction**

## W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Main training rectangles(2, 3), (4, 2), (6, 1), allcm:areas 6, 8, 6cm², mean 20/3. Validation(5, 2)area 10, centered 10/3cm². Picture trainingareas 2, 6, 10 mean6; query(2, 4)area 8 centers2; includingquerychangesmean 6.5 andfeature1.5. Mixedunitslength12 cm, width 30 mm=3 cmgivelength_over_width4, dimensionless. Measuredzero widthallowsarea 0 butratio unavailable; missing widthnotzero.

  Separate C1–C4 saved classifier comparison 3/4 both, oldmissC3, newfalsealarmC2; no total-accuracy gain or broad utility established.

- Reference for this exercise: open this week's student lesson, "Feature Engineering", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Compute the worked areas and their **training** mean.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 13

## W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Main training rectangles(2, 3), (4, 2), (6, 1), allcm:areas 6, 8, 6cm², mean 20/3. Validation(5, 2)area 10, centered 10/3cm². Picture trainingareas 2, 6, 10 mean6; query(2, 4)area 8 centers2; includingquerychangesmean 6.5 andfeature1.5. Mixedunitslength12 cm, width 30 mm=3 cmgivelength_over_width4, dimensionless. Measuredzero widthallowsarea 0 butratio unavailable; missing widthnotzero.

  Separate C1–C4 saved classifier comparison 3/4 both, oldmissC3, newfalsealarmC2; no total-accuracy gain or broad utility established.

- Center the validation area using the frozen **training** mean.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 14

## W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Main training rectangles(2, 3), (4, 2), (6, 1), allcm:areas 6, 8, 6cm², mean 20/3. Validation(5, 2)area 10, centered 10/3cm². Picture trainingareas 2, 6, 10 mean6; query(2, 4)area 8 centers2; includingquerychangesmean 6.5 andfeature1.5. Mixedunitslength12 cm, width 30 mm=3 cmgivelength_over_width4, dimensionless. Measuredzero widthallowsarea 0 butratio unavailable; missing widthnotzero.

  Separate C1–C4 saved classifier comparison 3/4 both, oldmissC3, newfalsealarmC2; no total-accuracy gain or broad utility established.

- Why is delivery delay invalid for a **prediction** made before dispatch?

> **My reasoning and evidence**

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MAIN LESSON · SECTION 16

# W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Main training rectangles(2, 3), (4, 2), (6, 1), allcm:areas 6, 8, 6cm², mean 20/3. Validation(5, 2)area 10, centered 10/3cm². Picture trainingareas 2, 6, 10 mean6; query(2, 4)area 8 centers2; includingquerychangesmean 6.5 andfeature1.5. Mixedunitslength12 cm, width 30 mm=3 cmgivelength_over_width4, dimensionless. Measuredzero widthallowsarea 0 butratio unavailable; missing widthnotzero.

  Separate C1–C4 saved classifier comparison 3/4 both, oldmissC3, newfalsealarmC2; no total-accuracy gain or broad utility established.

- How would you test adding area without claiming it must help?

**My reasoning and evidence**

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MAIN LESSON · SECTION 4

## Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Use the selected training areas 4, 8, 12cm²and new rectangle length 3 cm, width 4 cm. The average is fitted from training areas only; then apply it unchanged to the new area. This is a separate area/centering task, not the main 20/3 mean or pictured 6 mean. No prediction labels or model-utility result are supplied for this independent calculation.

- **Training** areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the **training** mean and centered new feature; state units.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Independent case: Apply independently

> **My independent explanation, trace and limits**
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> |---|
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> | |
> | |
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MAIN LESSON · SECTION 9

## Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Add only justified Plant Classifier features, with formula, source fields, expected benefit, and **leakage** test.

- The project change and **evidence** I will keep

**Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

# Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

MAIN LESSON · SECTION 10

# Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Use W2: Transform validation for **Feature engineering**. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

## Extra practice · Individual reasoning check

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Use the worked case for **Feature engineering**. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

## Reference · Earlier learning target

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

# Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Features are leaf length and width in cm; add length/width.
- Compute for8 and2; state boundary checks.
- Show your trace, calculation or **evidence**

**Write or draw your reasoning and evidence.**

▼ Extra practice · Changed case

REFERENCE · SECTION 7

## Extra practice · Changed case

> Keep this task’s complete inputs, stated rule and support record together.

### Task and evidence record

- A feature is leaf_length divided by leaf_width. A row has width zero. What should the pre**process**ing rule do before division?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

**Week 29 Test and reflect**

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

## Extra practice · Design a revealing test

> Keep this task’s complete inputs, stated rule and support record together.

### Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

## Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

- Record support used: none, hint or teacher explanation.

**My independently reasoned answer**

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

# Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- **Training** areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the **training** mean and centered new feature; state units.

**My reasoning and evidence**

▼ **Extension · Relationships and confounding — W1 Describe the evidence**

REFERENCE · SECTION 17

## Extension · Relationships and confounding — W1 Describe the evidence

> Retained extension task with its own values and teacher key.

### Extension question and evidence

- State the pattern in the three worked records without using **cause**d or guaranteed.

**My reasoning and evidence**

▼ **Extension · Relationships and confounding — W2 Identify a confounder**

REFERENCE · SECTION 18

## Extension · Relationships and confounding — W2 Identify a confounder

> Retained extension task with its own values and teacher key.

### Extension question and evidence

- Explain how prior experience could relate to both practice time and score.

**My reasoning and evidence**

▼ **Extension · Relationships and confounding — W3 Repair an overclaim**

REFERENCE · SECTION 19

# Extension · Relationships and confounding — W3 Repair an overclaim

> Retained extension task with its own values and teacher key.

## Extension question and evidence

- A learner predicts every student gains 2 points per extra hour. Explain why that is unsupported.

**My reasoning and evidence**

▼ Extension · Relationships and confounding — W4 Plan independently

REFERENCE · SECTION 20

# Extension · Relationships and confounding — W4 Plan independently

> Retained extension task with its own values and teacher key.

## Extension question and evidence

- A fictional dataset links more hints used with lower scores. Give a noncausal explanation and plan a fairer **comparison** of whether hints help.

> **My reasoning and evidence**

▼ Extension · Relationships and confounding — W5 State a remaining limit

REFERENCE · SECTION 21

# Extension · Relationships and confounding — W5 State a remaining limit

> Retained extension task with its own values and teacher key.

## Extension question and evidence

- Even a better **comparison** has only four fictional groups. What should be reported?

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 23

# Extra practice · Pictured case

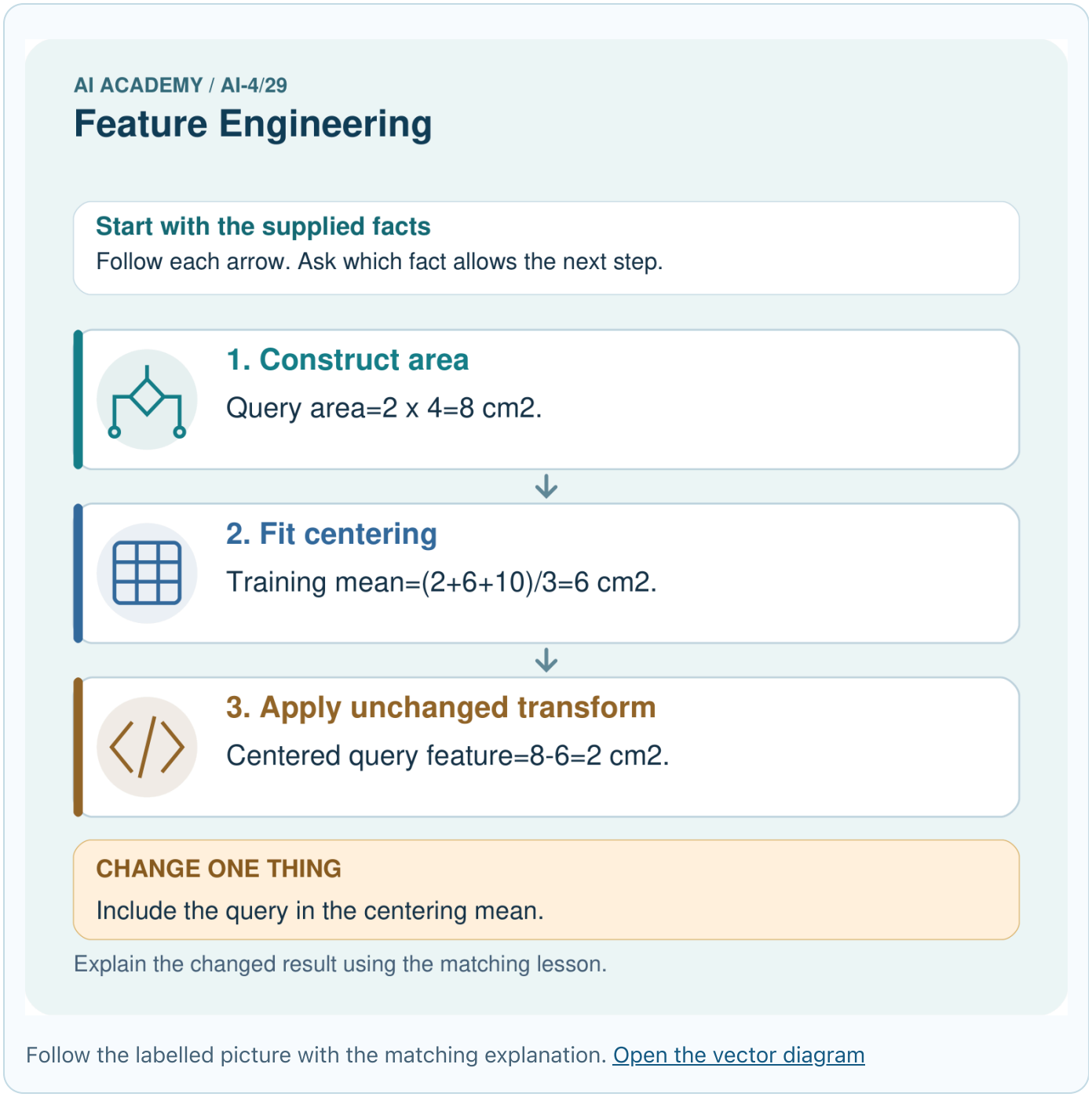
> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Engineer a feature with a **transformation** defined before evaluating a new leaf.
- **Training** rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.




Follow the labelled picture with the matching explanation. Open the vector diagram

**My picture-based explanation and changed-case prediction**

▼ **Reference · Project portfolio milestone**

REFERENCE · SECTION 24

# Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |

| Evidence to keep | My record |
| --- | --- |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 25

# Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

## First supported question and hint

- Engineer a feature with a **transformation** defined before evaluating a new leaf.
- Which labelled step matches this detail? Centered query feature=8-6=2 cm².
- 1. Construct area
- 2. Fit centering
- 3. Apply unchanged transform

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Apply unchanged transform Centered query feature=8-6=2 cm². Read the picture in order. Centered query feature=8-6=2 cm².

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

## Changed-case question and hint

- Changed-case practice: Include the query in the centering mean. Which conclusion is supported by the supplied facts?
- 1. Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

## Repair and keep the record

Review Learning Curves (ai-4/28). Goal: Read **training**-size learning curves. Short practice: What **comparison** is valid if the validation set changes between sizes?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised version?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

| | |
|---|---|
| | |
| | |
| | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Predict, run and debug**

**Keep your completed work**

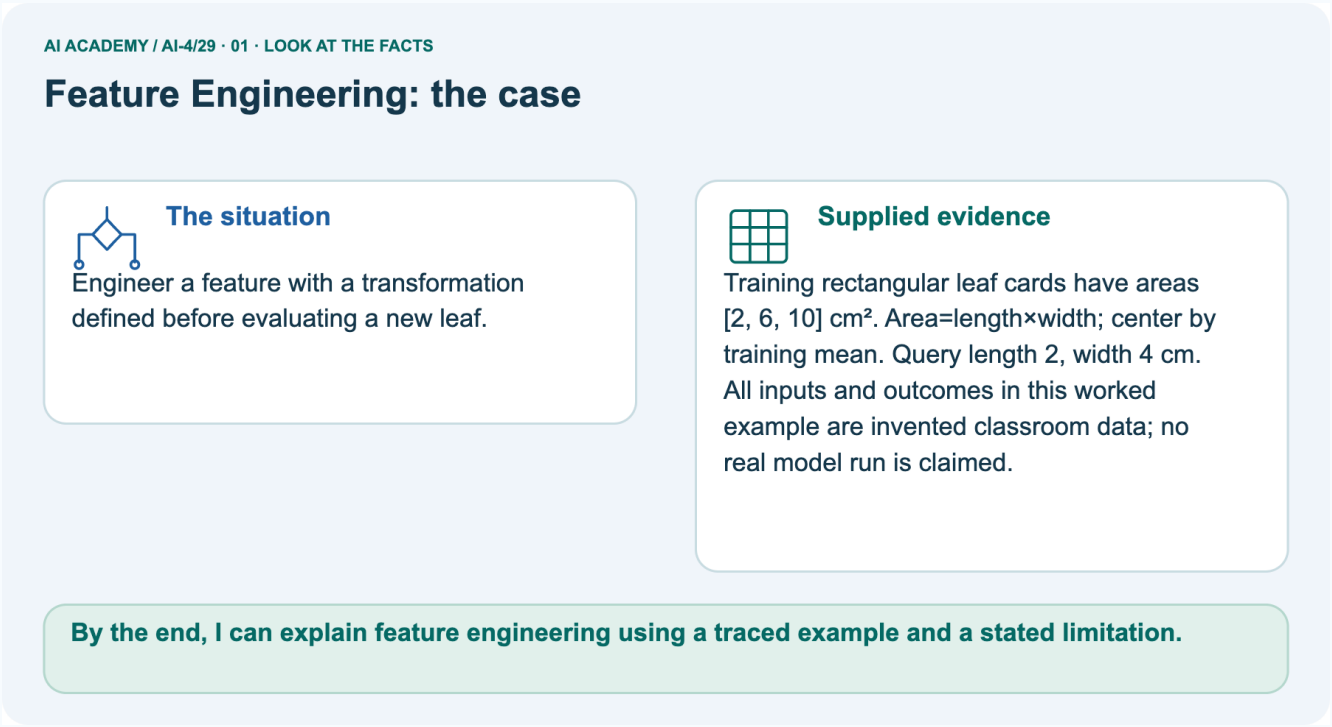
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 26

## Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

## 01. Feature Engineering: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

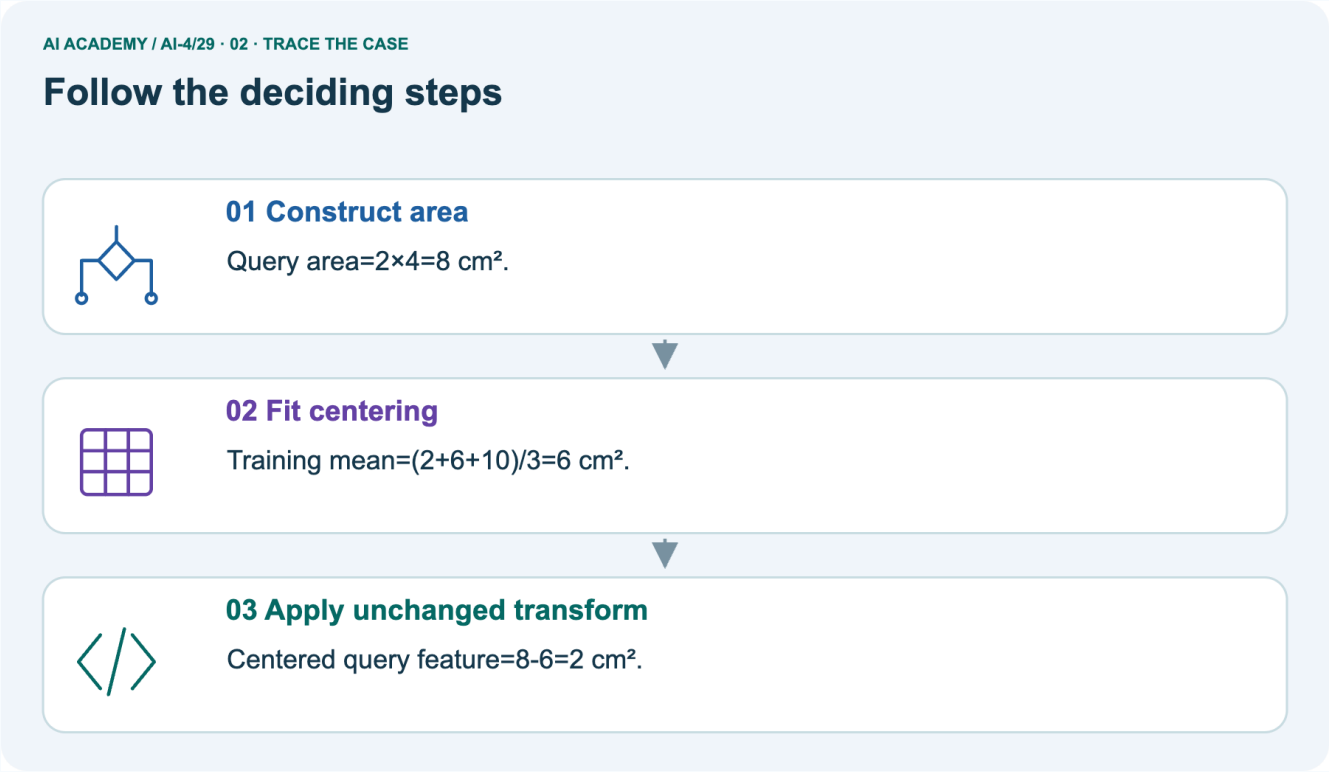
Engineer a feature with a **transformation** defined before evaluating a new leaf. Supplied facts: **Training** rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



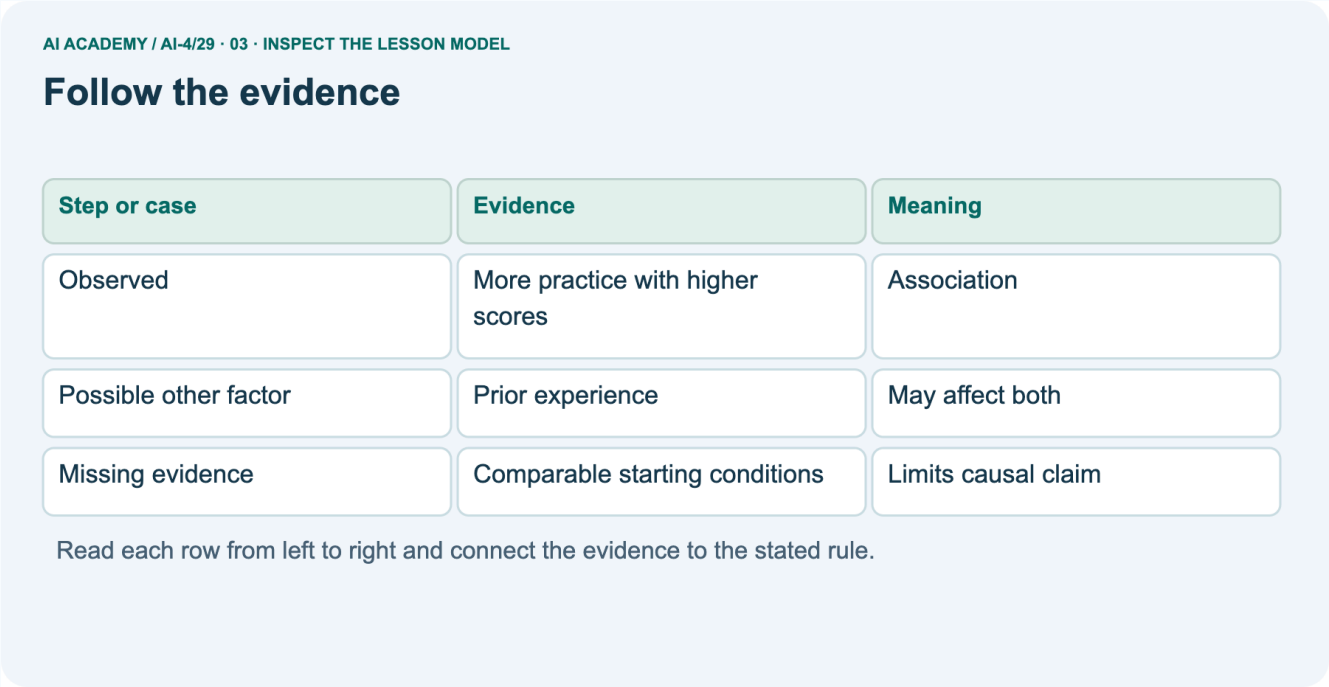

Follow the labelled picture with the matching explanation. Open the vector diagram

Construct area: Query area=2×4=8 cm². → Fit centering: **Training** mean=(2+6+10)/3=6 cm².
→ Apply unchanged transform: Centered query feature=8-6=2 cm².

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Follow the evidence



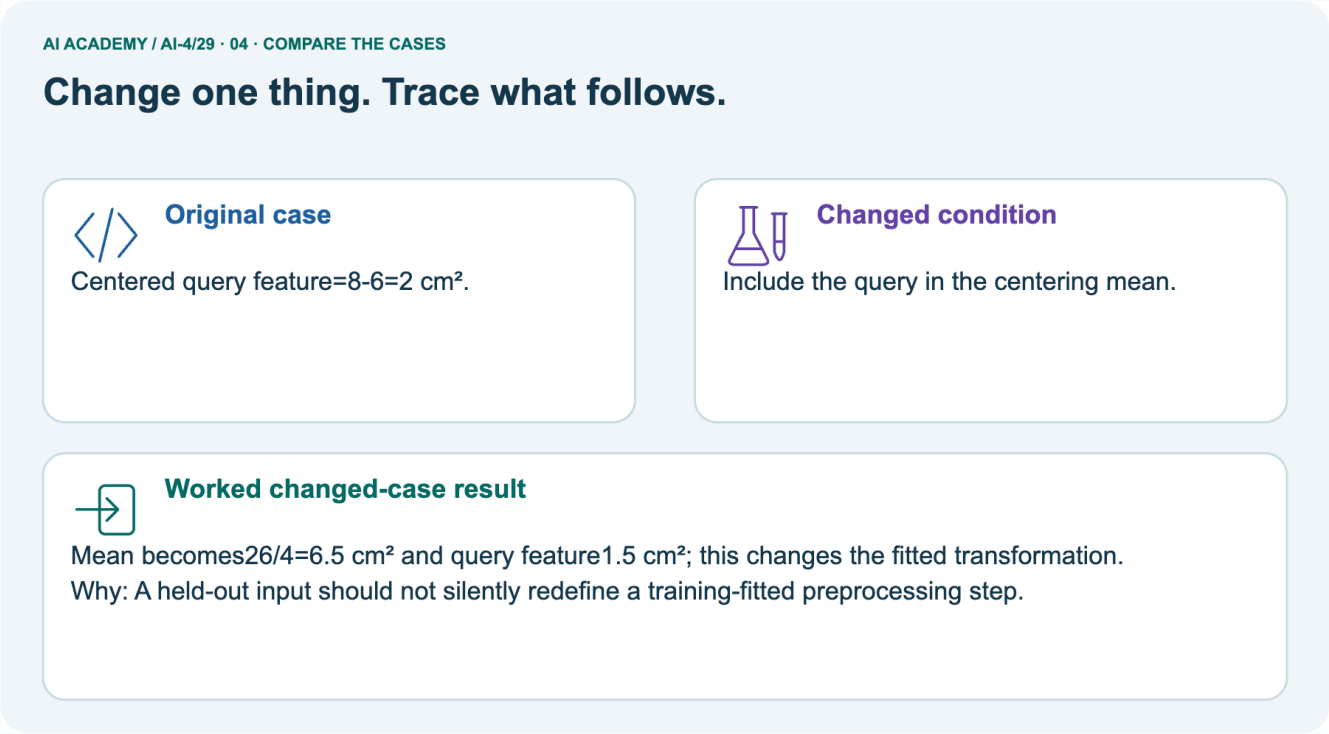

Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Observed | More practice with higher scores | **Association**. Possible other factor | Prior experience | May affect both. Missing **evidence** | Comparable starting conditions | Limits causal claim

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

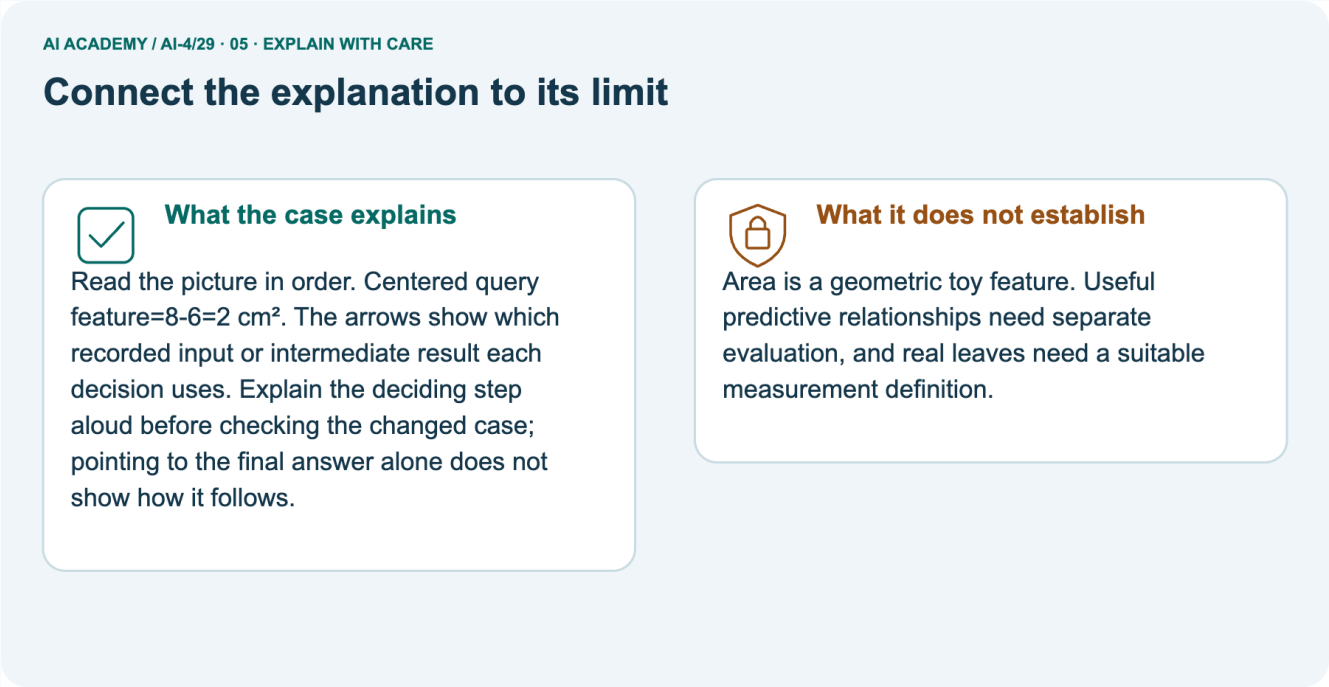
Original case: Centered query feature=8-6=2 cm². Changed condition: Include the query in the centering mean. Expected result: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Centered query feature=8-6=2 cm². The arrows show which recorded **input** or intermediate result each decision uses.

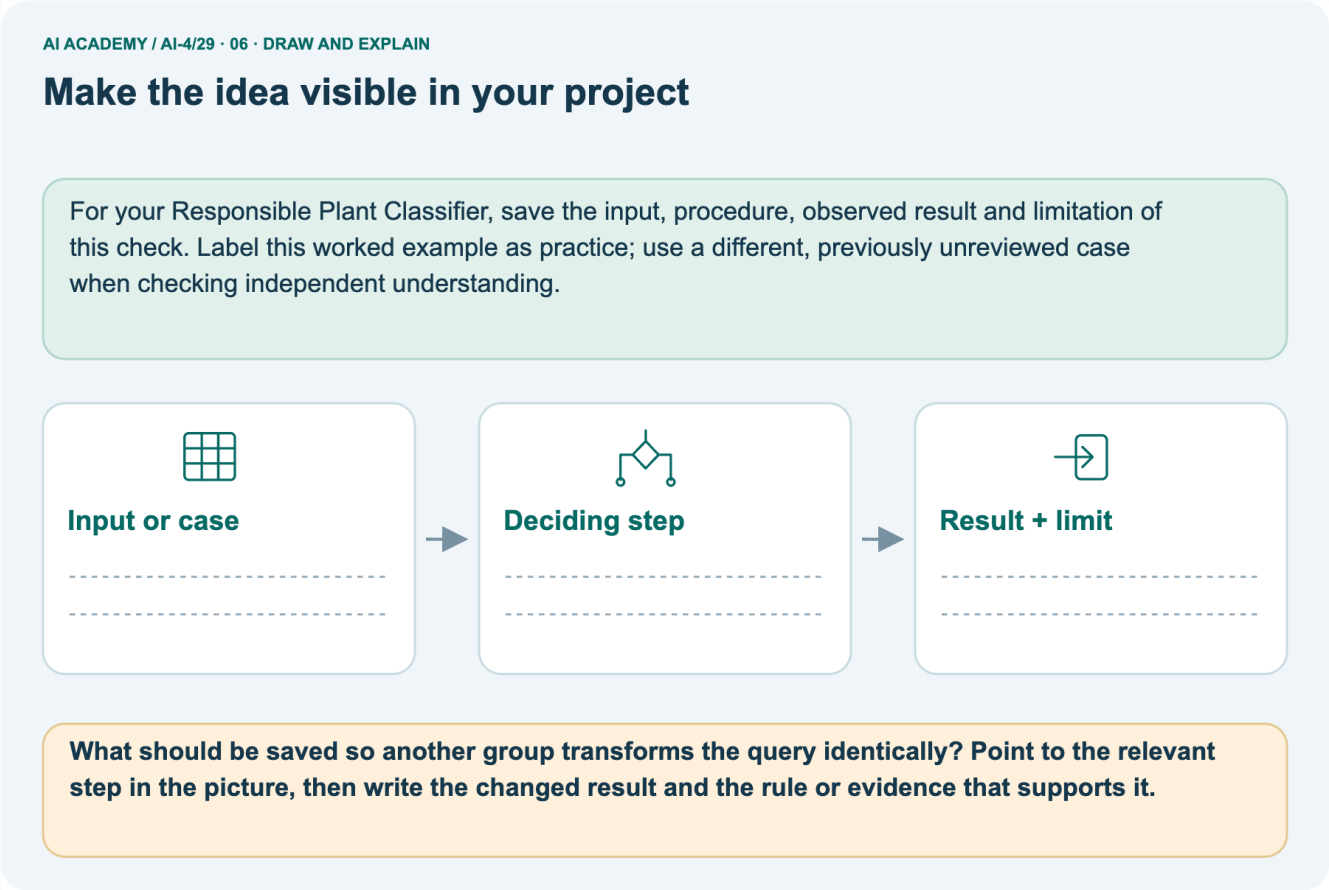
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Area is a geometric toy feature.

Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

**My picture explanation, deciding evidence and one limit**

▼ Earlier wording, alternatives and picture transcripts

**01. Feature Engineering: the case**

**02. Follow the deciding steps**

**03. Follow the evidence**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.